

Insurance coverage modifications

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Introduction

In insurance, very often we have to **transform the variables** that we are using. In general, we know that if X is a random variable, and we define $Y = g(X)$ where $g()$ is a monotone transformation, we have that

$$f_Y(y) = \left| \frac{d}{dy}(g^{-1}(y)) \right| f_X(g^{-1}(y)).$$

The most frequent transformations that we have to use in insurance are caused by **special policy coverage conditions**, specifically, the existence of **deductibles and limited loss** policy agreements.

Deductibles

Specifically, some transformations of the variables are necessary because of the existence of **special conditions in the coverage**, such as:

- the existence of a **maximum limit** in the coverage
- the existence of a **deductible**. Deductibles can be of two different types:
 - **absolute** (or ordinary): if the claim cost is higher than the deductible the insurer only pays the difference between the total claim cost and the deductible.
 - **relative**: if the claim cost is higher than the deductible, the insurer will pay the total claim cost.

Deductibles

The **excess loss** random variable Y^P (also called *per-payment variable*), is defined for the ordinary deductible case as:

$$Y^P = (X - d | X > d) = \begin{cases} \text{indefinido} & X \leq d \\ X - d & X > d \end{cases}$$

The **deductible** random variable Y^L (also called *per-loss variable*), is defined for the ordinary deductible case as:

$$Y^L = (X - d)_+ = \begin{cases} 0 & X \leq d \\ X - d & X > d \end{cases}$$

Deductibles: ordinary case

It is easy to proof that the density, distribution and survival functions of the **excess loss** random variable $Y^P = (X - d | X > d)$ with ordinary deductible are given by:

$$f_{Y^P}(y) = \frac{f_X(y + d)}{S_X(d)}, y > 0$$

$$F_{Y^P}(y) = \frac{F_X(y + d) - F_X(d)}{1 - F_X(d)}$$

$$S_{Y^P}(y) = \frac{S_X(y + d)}{S_X(d)}$$

Deductibles: ordinary case

The **deducible** random variable $Y^L = (X - d)_+$ for the ordinary case has a probability mass in 0 equal to $F_X(d)$ and for values higher than 0 has density function equal to:

$$f_{Y^L}(y) = f_X(y + d), y > 0$$

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$$F_{Y^L}(y) = F_X(y + d)$$

$$S_{Y^L}(y) = S_X(y + d)$$

Deductibles: relative case

The **excess loss** random variable Y^P (also called *per-payment variable*), is defined for the relative deductible case as:

$$Y^P = (X|X > d) = \begin{cases} \text{indefinido} & X \leq d \\ X & X > d \end{cases}$$

The **deductible** random variable Y^L (also called *per-loss variable*), is defined for the relative deductible case as:

$$Y^L = \begin{cases} 0 & X \leq d \\ X & X > d \end{cases}$$

Deductibles: relative case

The density, distribution and survival functions of the **excess loss** random variable Y^P is defined for the relative deductible case as,

$$f_{Y^P}(y) = \frac{f_X(y)}{1 - F_X(d)}, y > d$$

$$F_{Y^P}(y) = \frac{F_X(y) - F_X(d)}{1 - F_X(d)}$$

$$S_{Y^P}(y) = \frac{S_X(y)}{1 - F_X(d)}$$

Deductible: relative case

It is easy to prove that the density, distribution and survival functions of the **deductible** random variable Y^L is defined for the relative deductible case as,

$$f_{Y^L}(y) = \begin{cases} F_X(d) & y = 0 \\ f_X(y) & y > d \end{cases}$$

$$F_{Y^L}(y) = \begin{cases} F_X(d) & 0 \leq y \leq d \\ F_X(y) & y > d \end{cases}$$

$$S_{Y^L}(y) = \begin{cases} S_X(d) & 0 \leq y \leq d \\ S_X(y) & y > d \end{cases}$$

Limited loss

Similarly, we can also define the **limited loss** Y^U random variable as:

$$Y^U = X \wedge U = \begin{cases} X & X < U \\ U & X \geq U \end{cases}$$

and its distribution function is given by:

$$F_{Y^U}(y) = \begin{cases} F_X(y) & y < U \\ 1 & y = U \end{cases}$$

Limited loss

It is easy to prove that:

$$(X - d)_+ + (X \wedge d) = X$$

$$E(X \wedge d) = \int_{-\infty}^u xf(x)dx + u[1 - F(u)]$$

Moreover, for the ordinary deductible the following relationship is fulfilled:

$$E(X - d|X > d) = \frac{\int_d^{\infty} (x - d)f(x)dx}{1 - F(d)}$$

,

$$E[(X - d)_+] = E(X - d|X > d)(1 - F(d))$$

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